

hardware in the country is projected to increase from US\$ 45 billion in 2009 to US\$ 400 billion by 2020.

The estimated production will reach US\$ 104 billion by the year 2020, creating a gap of US\$ 296 billion in demand and production. This creates a unique opportunity for companies in the ESDM sector to look at India as their next destination to cater to the domestic Indian demand as well as act as an export hub.

Accordingly, the government had initiated several initiatives for the development of electronics sector in the country. The government has recently approved national policy on electronics (NPE). One of the important objectives of the NPE is to achieve a turnover of US\$ 400 billion by 2020, involving investment of about US\$ 100 billion and employment to around 28 million by 2020. This interalia includes achieving a turnover of US\$ 55 billion of chip design and embedded software industry, US\$ 80 billion of exports in the sector. Moreover, the policy also proposes setting up of more than 200 electronic manufacturing clusters. Another important objective of the policy is to significantly upscale high-end human resource creation to 2500 PhDs annually by 2020 in the sector.

As part of the efforts to promote the electronics sector, government of India would be setting up a semiconductor wafer fab in the country. India has become the hub for semiconductor design with nearly 2000 chips being designed per year, and more than 20,000 engineers are working in various aspects of chip design and verification. Annually, India is generating nearly US\$ 2 billion in revenues for the chip-design services. This provides an enabling environment for the semiconductor wafer fab, which will come up in India.

Several other policy initiatives have been approved in the last few months. These include providing very attractive financial investment in electronics manufacturing and preference to domestically-manufactured electronic goods in all government procurement as well as all those electronic goods

whose use has security implications for the country. The financial incentives are available not only for new units but also for units relocating from abroad. Besides, when we speak of electronics, apart from covering electronic hardware products relating to IT and office automation, telecom, consumer electronics, electronic components, etc, we cover a wide range of products relating to verticals like solar photovoltaic, automotive electronics, medical electronics, avionics, LED, etc. To address the issue of infrastructure, a scheme for the electronics manufacturing cluster provides 50 per cent of the cost of upgrading infrastructure and logistics as grant in aid from the government.

India has a vast pool of innovative and talented human resource. Our human resource is our strength that we offer to an investor from abroad. We produce over 200,000 engineers every year. The government of India also proposes to enhance the number of post graduates and PhDs in electronics, so that abundant manpower is available for the electronics industry to grow and flourish.

The ESDM industry in India has the following four key segments:

1. Electronic products
2. Electronics components
3. Semiconductor design services
4. Electronics manufacturing services (EMS)

Electronic products

The market for electronic products comprises domestic demand and products manufactured for exports. This market is expected to grow from US\$ 52.4 billion in 2013 to US\$ 74.6 billion in 2015, which includes exports of US\$ 10.5 billion in 2013 decreasing to US\$ 9.7 billion in 2015. Declining prices and increasing affordability are anticipated to propel domestic demand.

The total market is expected to grow from US\$ 41.91 billion to US\$ 64.85 billion in 2015. Total domestic manufacturing contributes only 35 per cent of this demand and the remaining 65 per cent is met through imported

products. The anticipated decline in the share of high-value-added manufacturing to 6.6 per cent of total market is a matter of serious concern. Timely policy intervention and incentives must be implemented urgently to increase the scope for local value-addition. Select products from the automotive and industrial electronics segments are the notable contributors to the high-value-addition activities within India, although their contribution to the total market of US\$ 64.85 billion is low. However, products with significant volumes and market size, like telecom, mobile devices, consumer electronics and IT/OA, suffer from lack of investment in product development and manufacturing.

Electronic components

The electronic components industry comprises semiconductors and passive components. As most electronic products are enabled by semiconductor components, it is important to analyse the overall consumption of semiconductors, which is expected to grow from US\$ 6.1 billion in 2011 to US\$ 9.66 billion in 2015. Although local demand and sourcing of semiconductors is limited to US\$ 2.94 billion in 2011, an increase to US\$ 3.69 billion in 2015 is due to import of electronic products and low domestic manufacturing. Continued innovation of semiconductor devices, coupled with their enhanced processing capability, has led to an increased consumption in mobile devices, information technology/office automation, telecom, automotive and industrial sectors.

Semiconductor design

Semiconductor design has traditionally been the strength for the Indian ESDM industry, generating revenues of US\$ 8.8 billion in 2011. There are more than 120 companies in India focused on semiconductor design for global products. This industry has witnessed a robust growth of 17.3 per cent since 2009 and today boasts of a 5.1 per cent share of the global pie. Considering the continued growth momentum, the